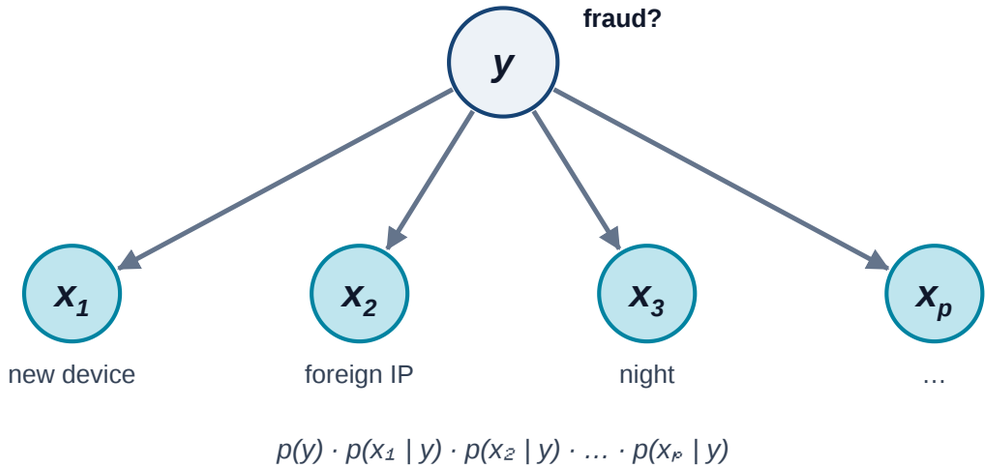
Probabilistic Machine Learning

Lecture 05: Bayesian Networks

Prof. Dr. Michael Granitzer

Where We Left Off



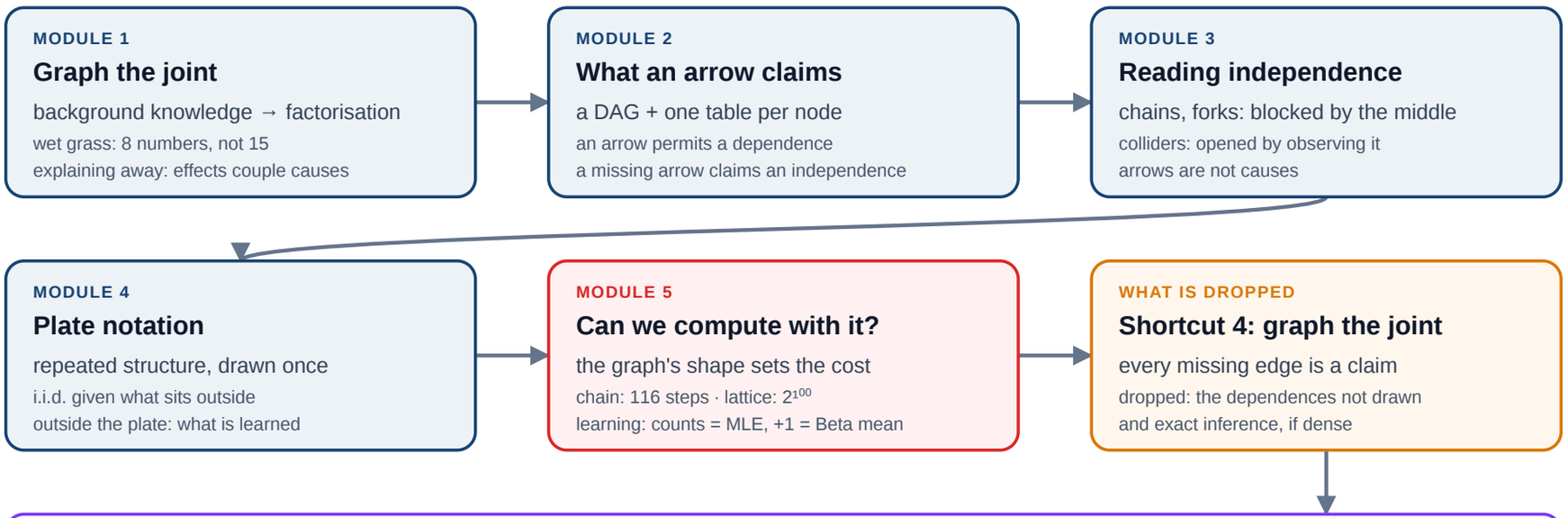
Lecture 03 made the joint affordable with one blanket assumption: **every** attribute independent of every other, given the class. It worked — and it counted correlated evidence twice.

What if we assumed only the independences we actually believe?

Shortcut 4 — graph the joint. Draw which variables depend on which; the graph *is* the factorisation. Dropped: every dependence **not drawn** — and, for a dense graph, affordable inference.

Argument Map

Lecture 05 · argument map



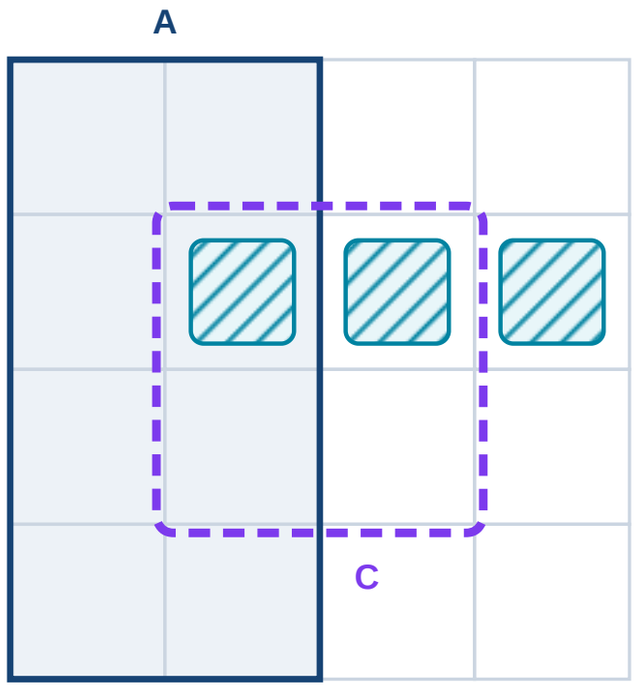
Course arc
ideal → shortcuts

L01	L02	L03	L04	**L05**	L06	L07
The ideal	The language	Peak · apart	Only p(y | x)	**Graph the joint**	Keep the loss	No shortcut

Graph the Joint

Guiding Question: Naive Bayes assumed every attribute independent of every other. What if we stated only the independences we believe — and what does conditioning then do to them?

Independent Given C — Counted



A 8 of 16 cells

B 3 of 16 cells

C 4 of 16 cells

Sixteen equally likely cells.

Dependent

$$P(A \cap B) = \frac{1}{16} \neq \frac{3}{32} = P(A)P(B)$$

Independent given C

$$P(A \cap B \mid C) = \frac{1}{4} = P(A \mid C)P(B \mid C)$$

Why Is the Grass Wet?

Tracey sees her lawn is wet. **Rain — or did she leave the sprinkler on?** Then she sees that her neighbour Jack's lawn is wet too.

variable	meaning (1 = yes)
R	it rained
S	Tracey's sprinkler was on
J	Jack's grass is wet
T	Tracey's grass is wet

Before any arithmetic: does Jack's wet lawn make the sprinkler **more** or **less** likely?

Background Knowledge as Independence

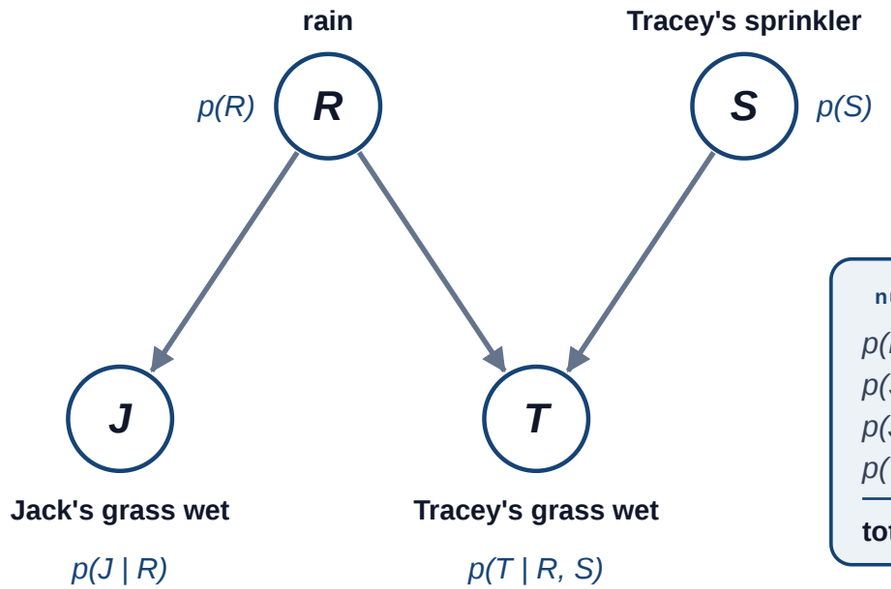
The joint needs $2^4 - 1 = 15$ numbers. The chain rule rewrites it without shrinking it:

$$p(T, J, R, S) = p(T \mid J, R, S) \cdot p(J \mid R, S) \cdot p(R \mid S) \cdot p(S)$$

- → Tracey's lawn ignores Jack's, given R and S: $p(T \mid J, R, S) = p(T \mid R, S)$
- → Jack's lawn ignores Tracey's sprinkler: $p(J \mid R, S) = p(J \mid R)$
- → the sprinkler does not change the weather: $p(R \mid S) = p(R)$

> $$p(T, J, R, S) = p(T \mid R, S) \cdot p(J \mid R) \cdot p(R) \cdot p(S)$$

The Graph Is the Factorisation



numbers needed	
p(R)	**1**
p(S)	**1**
p(J | R)	**2**
p(T | R, S)	**4**
total	**8**

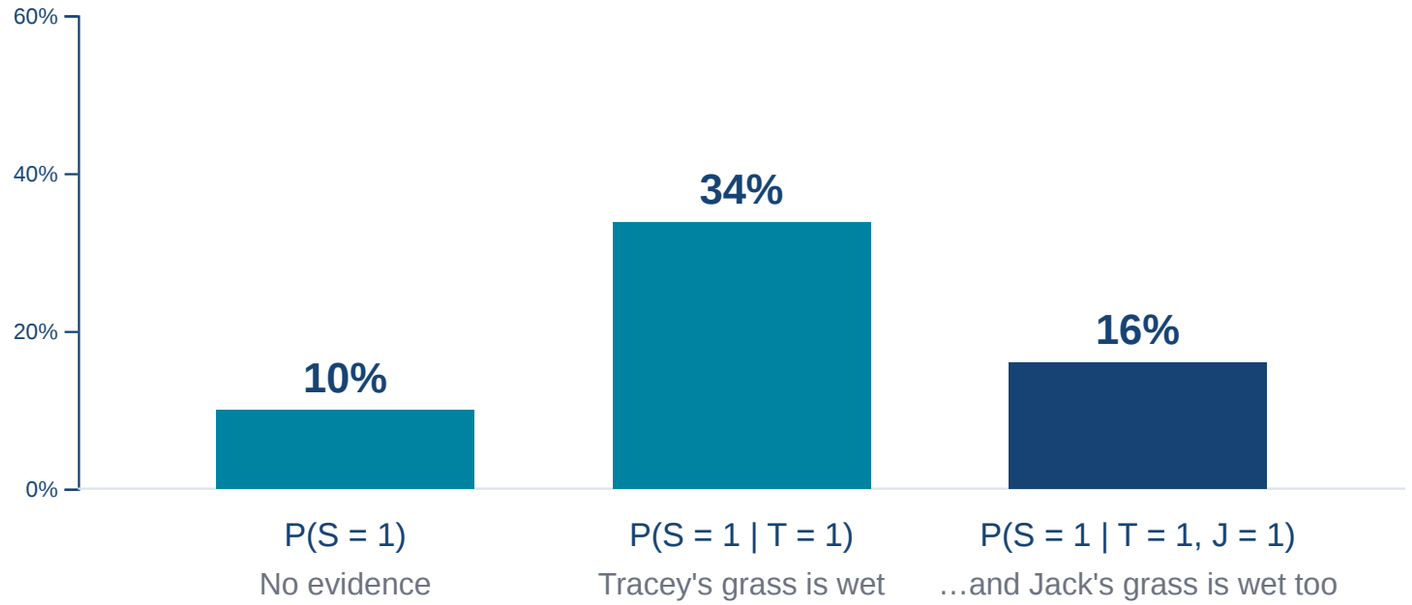
full joint: 15

factor	$= 1$
$p(R = 1)$	0.2
$p(S = 1)$	0.1
$p(J = 1 \mid R = 0)$	0.2
$p(J = 1 \mid R = 1)$	1.0
$p(T = 1 \mid R = 0, S = 0)$	0.0
$p(T = 1 \mid R = 0, S = 1)$	0.9
$p(T = 1 \mid R = 1, S = 0)$	1.0
$p(T = 1 \mid R = 1, S = 1)$	1.0

One node per variable, one arrow per conditioning bar, one table per node.

What Does the Evidence Say?

Rain explains the wetness — the sprinkler falls back towards its prior



Explaining Away

$$p(S=1 \mid T=1, J=1) = \frac{\sum_R p(T{=}1 \mid R, S{=}1)p(J{=}1 \mid R)p(R)p(S{=}1)}{\sum_{R,S} p(T{=}1 \mid R, S)p(J{=}1 \mid R)p(R)p(S)} = \frac{0.0344}{0.2144} \approx 0.16$$

Rain and sprinkler are independent: $R \perp\!\!\!\perp S$. Once their **common effect** T is observed they compete to explain it, and evidence for one lowers the other. Conditioning on a common cause removes a dependence; conditioning on a common effect **creates** one.

Mini-Exercise

Tracey's lawn is wet. Then she finds the **sprinkler was on**.

Does rain become more or less likely than it was a moment ago?

Mini-Exercise

Tracey's lawn is wet. Then she finds the **sprinkler was on**.

Does rain become more or less likely than it was a moment ago?

Less. The sprinkler explains the wet lawn. $P(R = 1 \mid T = 1) \approx 0.74$ falls to $P(R = 1 \mid T = 1, S = 1) = 0.02/0.092 \approx 0.22$ — almost back to the prior 0.2. Explaining away works in both directions.

Takeaway

Graph the joint: background knowledge becomes conditional independence, and the graph is a picture of how the joint splits — 8 numbers instead of 15. Every *missing* edge is a claim; conditioning can make variables independent (common cause) or dependent (common effect).

What an Arrow Claims

Guiding Question: What exactly does an arrow in a probabilistic model claim — and what does a *missing* arrow claim?

Belief Networks

Bayesian belief network. A **directed acyclic graph** whose nodes are random variables, with one conditional table per node given its parents:

$$p(x_1, \ldots, x_n) = \prod_{i=1}^{n} p\big(x_i \mid \mathrm{pa}(x_i)\big)$$

An arrow permits a dependence. A missing arrow claims an independence: each variable is independent of its non-descendants, given its parents (the **local Markov property**).

Which independence does the wet-grass graph claim for Jack's lawn J?

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Which independence does the wet-grass graph claim for Jack's lawn J?

$J \perp\!\!\!\perp \{S, T\} \mid R$: once we know whether it rained, nothing else says more about Jack's lawn.

Counting the Numbers

model, binary variables	full joint	graph
wet grass, 4 nodes	15	$1 + 1 + 2 + 4 = 8$
naive Bayes, class + 30 flags	$2^{31} - 1 \approx 2.1 \cdot 10^9$	$1 + 2 \cdot 30 = 61$
20 nodes, at most 3 parents each	$2^{20} - 1 \approx 10^6$	at most $20 \cdot 2^3 = 160$

Where does the saving come from — and what makes it disappear?

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Where does the saving come from — and what makes it disappear?

A node with k binary parents needs 2^k numbers. The saving comes from **missing arrows**; it disappears as soon as some node has many parents.

Keeping Tables Small: Noisy-OR

A symptom y with many possible causes $x_1, \ldots, x_k$: **each present cause fires on its own** with probability λ_i, and y occurs if any fires:

$$p(y = 0 \mid x_1, \ldots, x_k) = \prod_{i : x_i = 1} (1 - \lambda_i)$$

k numbers instead of 2^k. Two causes present, $\lambda = 0.8$ and 0.6: $p(y = 1) = 1 - 0.2 \cdot 0.4 = 0.92$.

THE ASSUMPTION BOUGHT

Causes act **independently** — no interaction. Two drugs that are harmless alone but dangerous together break it.

Divorcing is the other trick: insert a summary node ("total load") between a group of parents and the child.

Mini-Exercise

Tracey and Jack share a garden hose: add the arrow $S \to J$ to the wet-grass network.

How many numbers does the network need now — and which independence claim is gone?

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Tracey and Jack share a garden hose: add the arrow $S \to J$ to the wet-grass network.

How many numbers does the network need now — and which independence claim is gone?

10. $p(J \mid R, S)$ needs 4 numbers instead of 2: $1 + 1 + 4 + 4$. Gone is $J \perp\!\!\!\perp S \mid R$ — the missing arrow was that claim.

Takeaway

A belief network is **a DAG plus one table per node**. An arrow only permits a dependence; every **missing** arrow claims an independence, and each claim saves numbers. Shrinking tables further — noisy-OR, divorcing — buys more savings with more assumptions, which should be said out loud.

Reading Independence off the Graph

Guiding Question: Two causes are independent until you observe their shared effect. Can we see such things in the picture — before computing a single number?

Three Ways Three Nodes Connect

CHAIN

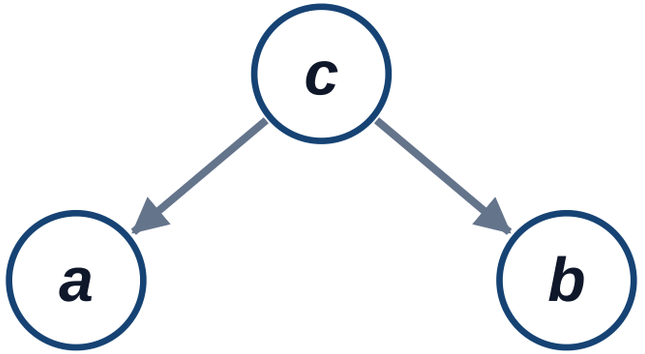


a, b: **dependent**

given c: **independent**

observing c blocks the path

FORK · COMMON CAUSE

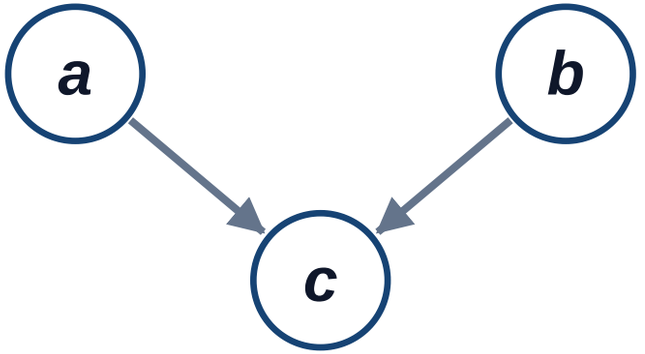


a, b: **dependent**

given c: **independent**

observing c blocks the path

COLLIDER · COMMON EFFECT



a, b: **independent**

given c: **dependent**

observing c opens the path

Every independence question in a DAG reduces to these three.

d-Separation

A path between X and Y is **blocked** by a set Z if it contains

- a chain or fork whose middle node **is in** Z, or
- a collider whose middle node — and every descendant of it — is **not in** Z.

If every path is blocked, X and Y are **d-separated** by Z, and then $X \perp\!\!\!\perp Y \mid Z$ in every distribution that factorises over the graph.

Sound, not complete: a particular table may add independences the graph does not show — in our tables, once it rains, Tracey's lawn ignores the sprinkler.

d-Separation on the Wet Grass

The only path from J to S: $J \leftarrow R \rightarrow T \leftarrow S$ — a fork at R, a collider at T.

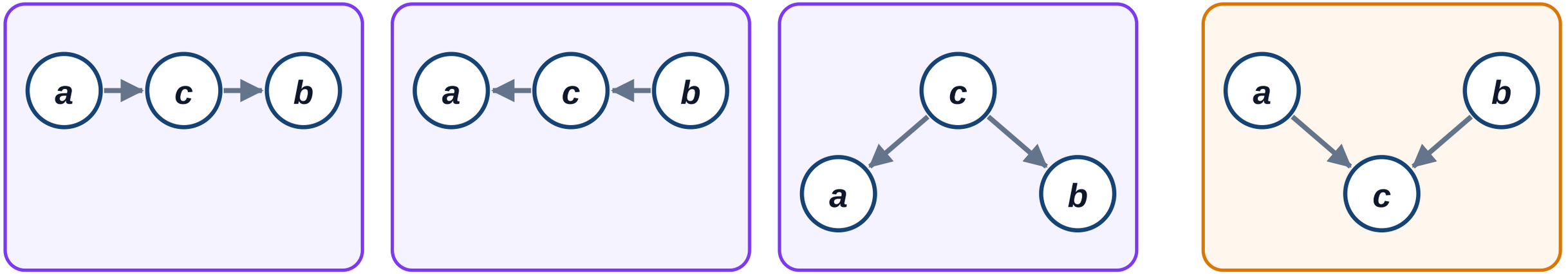
query	path	verdict	the numbers
$J \perp\!\!\!\perp S$?	collider T unobserved: blocked	independent	$P(J{=}1) = 0.36 = P(J{=}1 \mid S{=}1)$
$J \perp\!\!\!\perp S \mid T$?	collider T observed: open	dependent	$P(J{=}1 \mid T{=}1) = 0.79$, but 0.37 if also $S{=}1$
$J \perp\!\!\!\perp S \mid T, R$?	fork R observed: blocked	independent	$P(S{=}1 \mid T{=}1, R{=}1) = 0.1$, with or without $J{=}1$

> The middle row is Module 1's explaining away: $10\% \rightarrow 34\% \rightarrow 16\%$ for the sprinkler. The graph predicted it before any number was computed.

Arrows Are Not Causes

same claim: a, b independent given c

different claim



data cannot tell these three apart

the direction of the arrows is not in the data

a, b independent;
dependent given c

Same skeleton, same colliders: **the same independences** (Markov equivalence). Data cannot orient such arrows — they are **not** causal claims.

Mini-Exercise

Tracey's doormat gets wet whenever her lawn is: add $T \rightarrow D$. You see only a **wet doormat —** not the lawn.

Are rain and sprinkler still independent?

Mini-Exercise

Tracey's doormat gets wet whenever her lawn is: add $T \to D$. You see only a **wet doormat** — not the lawn.

Are rain and sprinkler still independent?

No. D is a descendant of the collider T, so observing it opens the path $R \to T \leftarrow S$: a wet doormat also makes rain and sprinkler compete to explain it.

Takeaway

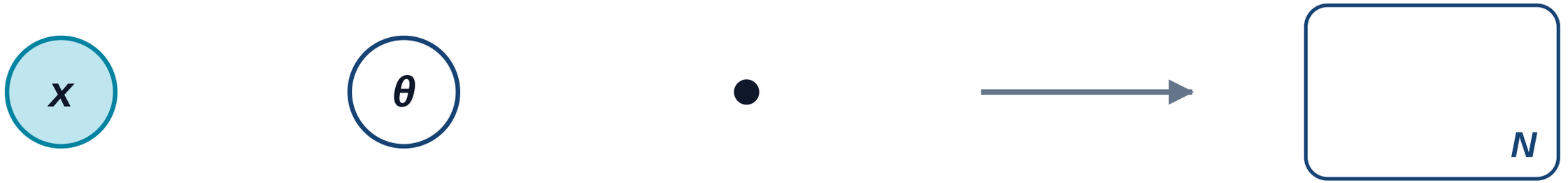
Independence in a belief network is a **property of the graph**: chains and forks are blocked by observing the middle, colliders are opened by it (or by a descendant). d-separation applies that to every path. What it cannot read off is the direction of cause: equivalent graphs make the same claims.

Drawing Repeated Structure

Guiding Question: A model of 10,000 observations should not need 10,000 nodes. How is a repeated structure drawn — and what does the drawing claim about the observations?

Drawing a Model

A probabilistic model is a factorisation, and the diagram is its picture. Five conventions carry almost all of it:



shaded
observed: in the data

open
latent or parameter

solid dot
fixed, not inferred

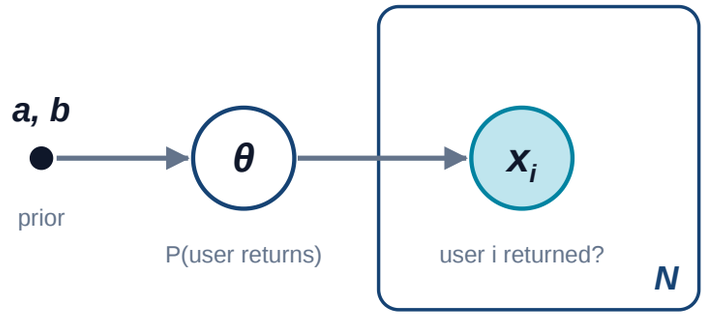
arrow
child conditioned on parent

plate
repeat N times

> Nothing new is claimed: the arrows are read exactly as in a belief network. The shading only says **what the data fixes** — which variables we condition on.

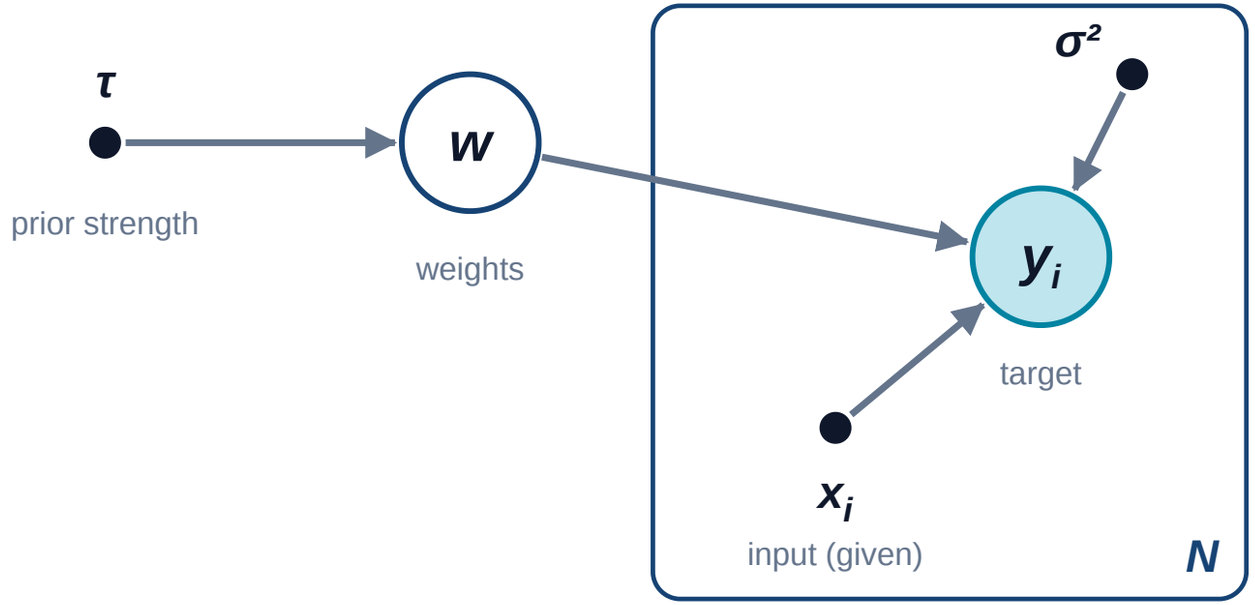
Plates

A **plate** is a box with a count in its corner: *repeat what is inside N times*. A model of 10,000 users does not need 10,000 nodes.
The copies are **independent given what sits outside** — the parameters, which learning estimates. Nested plates draw grouped data: K hospitals with N_k patients each.



> Given θ the users are independent. With θ unknown they are **not**: in Lecture 02, the first ten users changed our prediction for the eleventh. That dependence is exactly why learning from data works.

Reading a Diagram



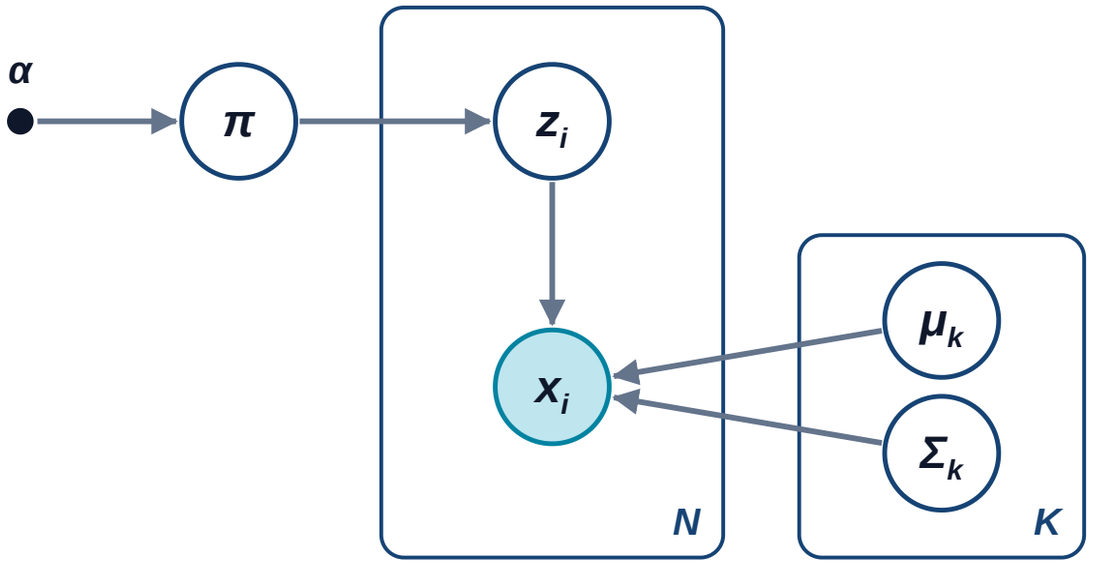
Read off the factorisation:

$$p(w \mid \tau) \prod_{i=1}^{N} p(y_i \mid x_i, w, \sigma^2)$$

The plate became the product; each arrow became a conditioning bar.

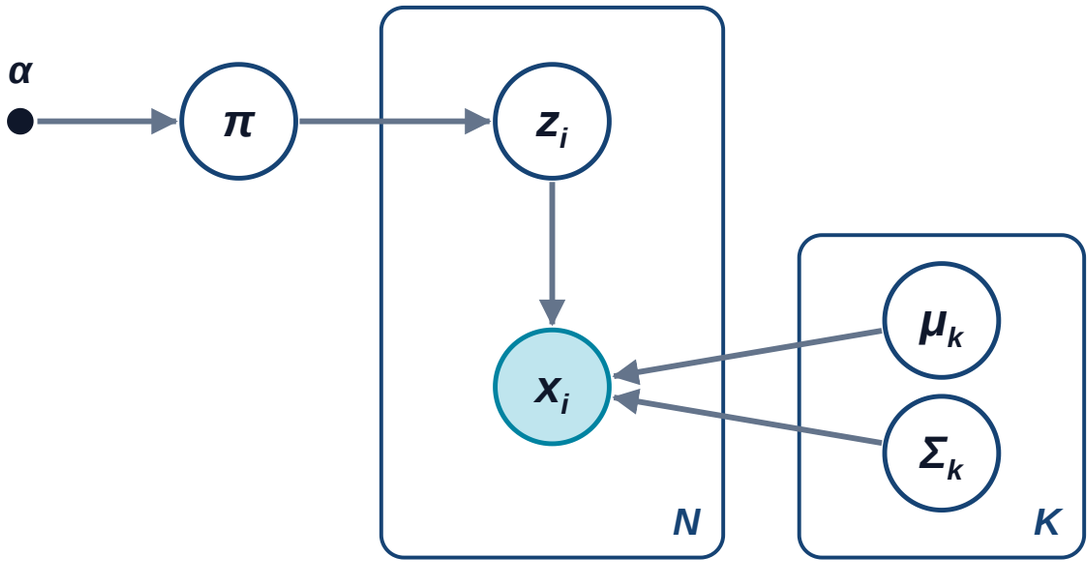
x_i is a **dot**: the inputs are given, not modelled — this is a discriminative model, $p(y \mid x)$, with a prior on w. Lecture 07 conditions on the shaded y_i and keeps the whole posterior over w.

Mini-Exercise



Which nodes does learning have to infer — and why is α not among them?

Mini-Exercise



Which nodes does learning have to infer — and why is α not among them?

π, every μ_k, Σ_k and every label z_i — all **open**.

α is a **dot**: fixed by us. Only the x_i are observed:

$$p(\pi \mid \alpha) \prod_k p(\mu_k, \Sigma_k) \prod_i p(z_i \mid \pi) p(x_i \mid z_i, \mu, \Sigma)$$

A **mixture of Gaussians** — Lecture 07.

Takeaway

A model diagram is a factorisation you can see: **shading** marks what the data fixes, **plates** mark what repeats independently *given* what sits outside, and **what sits outside** is what learning has to estimate.

Can We Compute With It?

Guiding Question: We can now write the joint down. Can we actually condition on it — and where do its numbers come from?

Three Questions You Can Ask a Model

question	on the wet grass	with complete data
infer unobserved variables	$P(S \mid T, J)$	cost set by the graph's shape
learn the parameters	the 8 table entries	counting
learn the structure	which arrows at all?	hardest — and only up to equivalence

Inference by Elimination

Explaining away needed $P(T{=}1, J{=}1)$. Sum the factorised joint — and **push each sum as far right as it goes**:

$$P(T{=}1, J{=}1) = \sum_R p(R)p(J{=}1 \mid R) \underbrace{\sum_S p(S)p(T{=}1 \mid R, S)}_{m(R)}$$

Inner sum first

$m(1) = 0.1 \cdot 1 + 0.9 \cdot 1 = 1$

$m(0) = 0.1 \cdot 0.9 + 0.9 \cdot 0 = 0.09$

Then the outer one

$0.2 \cdot 1 \cdot 1 + 0.8 \cdot 0.2 \cdot 0.09$

$= 0.2 + 0.0144 = 0.2144$ ✓

Why Pushing Sums In Pays

A chain of n binary variables $X_1 \to X_2 \to \cdots \to X_n$. To get $p(x_n)$:

$$2^{n-1}$$

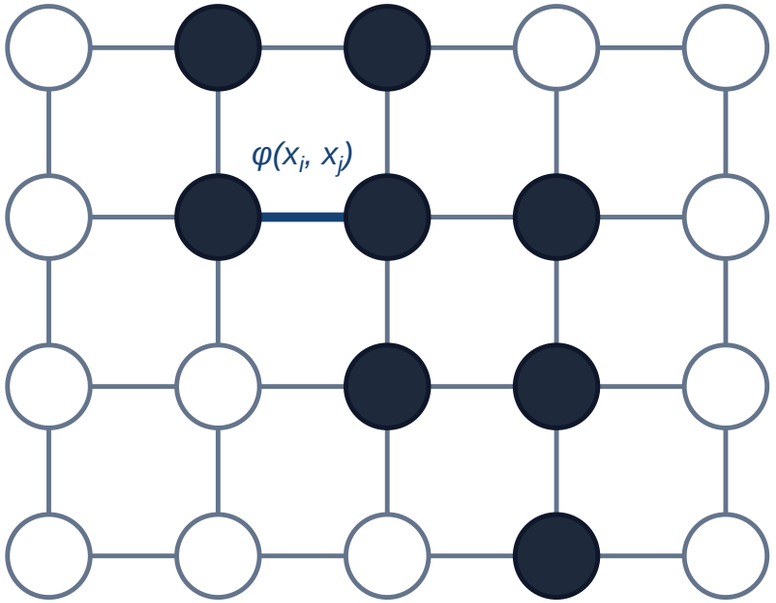
terms if every other variable is summed at once — for $n = 30$, about 537 million

$$4(n-1)$$

operations if the sums are pushed in, one variable at a time — for $n = 30$, 116

Eliminating X_1, then X_2, … passes a **two-entry table** along the chain. The cost is set by the **largest table** created on the way — by the graph's shape, not its size.

Undirected Models



each edge: a potential that prefers neighbours to agree

Pixels influence each other **symmetrically**. A **Markov network** puts a non-negative **potential** ϕ on each edge and normalises once, globally:

$$p(x) = \frac{1}{Z} \prod_{(i,j)} \phi(x_i, x_j), \qquad Z = \sum_{x} \prod_{(i,j)} \phi(x_i, x_j)$$

Two pixels, $\phi = 3$ if equal, 1 if not: $Z = 3 + 1 + 1 + 3 = 8$, so $P(\text{equal}) = 6/8$.

Directed and Undirected

	belief network	Markov network
edges	directed, acyclic	undirected
factors	conditional tables, normalised locally	potentials, one global Z
natural for	causal or generative stories	symmetric interaction: pixels, neighbours
observing a common effect	creates dependence	— (no colliders)

Inference runs on the undirected form: **moralisation** joins parents that share a child (R – S in the wet grass) and drops the arrow heads; elimination works on that graph.

Writable Is Not Queryable

A 100×100 black-and-white image as a pixel lattice:

79,200

numbers to **write it down**: 19,800 edges, 4 per potential

$$2^{100} \approx 10^{30}$$

entries in the table that elimination must build — a whole row of pixels at once

Exact inference in belief networks is **NP-hard** in general, and so is approximate inference with a guaranteed error bound. What remains: exploit the shape where it is kind (chains, trees), and otherwise **sample** or **approximate** — trading the guarantee for an answer.

Learning the Parameters Is Counting

Fifty observed days. Estimate Jack's table $p(J{=}1 \mid R)$:

	days	Jack's lawn wet	MLE (peak)	$+1$ pseudo-counts (mean)
$R = 0$	40	8	$8/40 = 0.20$	$9/42 \approx 0.21$
$R = 1$	10	10	$10/10 = 1.0$	$11/12 \approx 0.92$

> Counting **is** maximum likelihood (L03); pseudo-counts **are** the Beta posterior mean (L02) — Laplace smoothing (L03). "Rain always wets the grass" was an MLE on ten days.

Hidden variables leave no counts: infer them, re-count, repeat — EM, Lecture 07.

Learning the Structure

Score and search

Try graphs, score each by how well it fits the data *per parameter*, keep the best. The number of DAGs grows super-exponentially.

Test independences

Run conditional-independence tests and keep the graphs consistent with them — d-separation, read backwards.

Either way, data determines a graph only **up to Markov equivalence**: arrows between equivalent graphs are not in the data. A learned structure is not a causal discovery.

Mini-Exercise

Jack's lawn was wet on all **4** rainy days in a small sample.

What does counting give for $p(J{=}1 \mid R{=}1)$ — and what does one pseudo-count on each side give?

Mini-Exercise

Jack's lawn was wet on all **4** rainy days in a small sample.

What does counting give for $p(J{=}1 \mid R{=}1)$ — and what does one pseudo-count on each side give?

Counting: $4/4 = 1$ — "rain always wets the lawn", from four days. **Smoothed:** $5/6 \approx 0.83$, the posterior mean under $\mathrm{Beta}(1,1)$. Same data, two readings: the peak and the mean.

Takeaway

A graphical model makes the joint **writable**; its **shape** decides whether it is **queryable** — elimination is cheap on chains and trees and hopeless on a lattice. With complete data the parameters are counts: the MLE, or with pseudo-counts the Beta posterior mean. The structure is learnable only up to equivalence.

Appendix

Uncertain and Unreliable Evidence

Uncertain (soft) evidence

We are not sure of the observation itself: a distribution $\tilde{x}$ over its states. Sum over them:

$$p(y \mid \tilde{x}) = \sum_{x} p(y \mid x) p(x \mid \tilde{x})$$

We look at the lawn from the window without touching it: $p(T{=}1 \mid \tilde{T}) = 0.8$.

Unreliable evidence

We keep the report but **replace** its table by our own judgement of how reliable it is:

$$p(x \mid \mathrm{pa}(x)) \to \tilde{p}(x \mid \mathrm{pa}(x))$$

Jack says his lawn is wet, but he did not check.

Both are drawn with a **dashed** edge.